

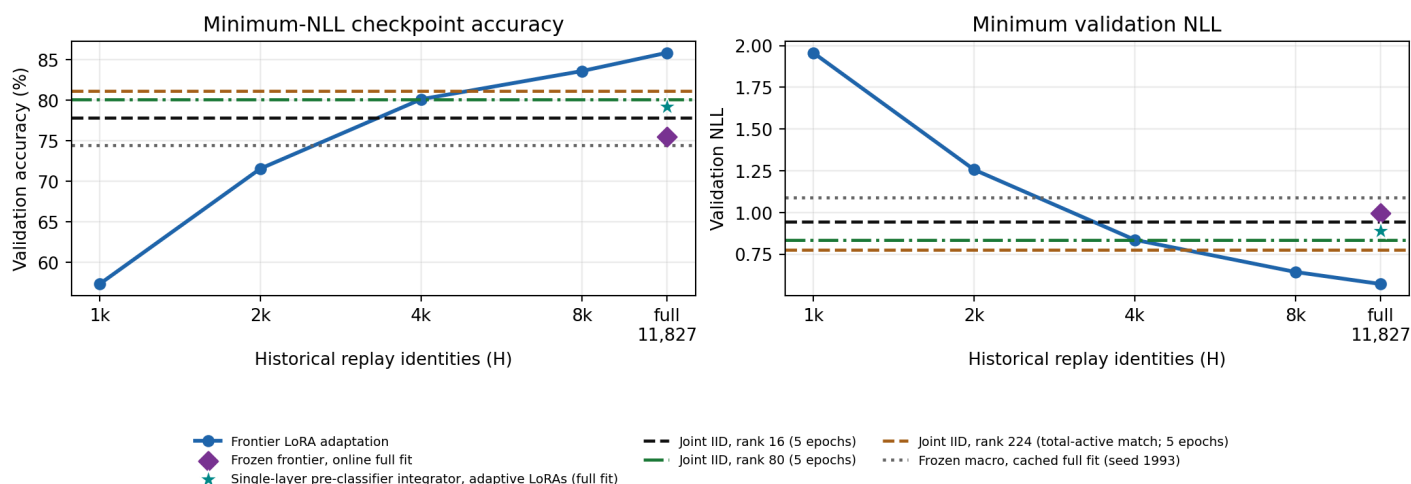
ImageNet-R stage-31 frontier-LoRA adaptation

Can jointly adapting five fragmented node representations close the same-split joint-IID accuracy and NLL gap?

Selected-checkpoint result. The minimum-NLL adaptive condition is Frontier LoRA adaptation (H=11,827; full fit): **85.864% accuracy / 0.5710 NLL**. That is 8.003 percentage points higher and 0.3715 NLL lower than the five-epoch rank-16 joint-IID reference. Frontier LoRA adaptation (H=4,096) is the smallest H to reach both rank-16 joint-IID metrics, first doing so at epoch 6.

Integrator architecture ablation. The single-layer head reaches **79.239% / 0.8907 NLL** at its selected checkpoint (-6.625 accuracy points and +0.3197 NLL versus selected macro full history).

Selected frontier checkpoints versus five-epoch joint IID



Blue macro-frontier points and the teal single-layer point use each condition's minimum-NLL checkpoint. Black rank-16, green rank-80, and ochre rank-224 joint-IID lines are fixed epoch-five endpoints.

Matched-exposure architecture and capacity comparison

Condition	LoRA layout	LoRA parameters	Other trainable parameters	Total active parameters	Epoch-5 accuracy	Epoch-5 NLL
Joint IID, rank 16 (5 epochs)	1 x 16	1,327,104	95,356	1,422,460	77.862%	0.9425
Joint IID, rank 80 (5 epochs)	1 x 80	6,635,520	95,356	6,730,876	80.125%	0.8339
Joint IID, rank 224 (total-active match; 5 epochs)	1 x 224	18,579,456	95,356	18,674,812	81.109%	0.7772
Single-layer pre-classifier, adaptive LoRAs (epoch 5)	5 x 16	6,635,520	768,200	7,403,720	78.649%	0.9560
Adaptive frontier, full history (epoch 5)	5 x 16	6,635,520	12,055,496	18,691,016	84.880%	0.5897

Every row uses exactly five complete passes over the 12,194-image fit population. "Other" means the trainable joint classifier, linear integrator, or macro integrator; frozen parameters are omitted. Rank 80 matches aggregate node-LoRA capacity. Rank 224 is the closest total-active-parameter match.

Complete condition summary

Condition	Min-NLL epoch	Accuracy at min NLL	Minimum NLL	Maximum accuracy	NLL at max accuracy	Runtime
Frontier LoRA adaptation (H=1,024)	31	57.330%	1.9572	57.527%	1.9624	21.9 min
Frontier LoRA adaptation (H=2,048)	25	71.564%	1.2565	71.564%	1.2565	30.1 min
Frontier LoRA adaptation (H=4,096)	12	80.125%	0.8360	80.846%	0.8443	47.1 min
Frontier LoRA adaptation (H=8,192)	8	83.601%	0.6430	84.093%	0.6534	88.2 min
Frontier LoRA adaptation (H=11,827; full fit)	13	85.864%	0.5710	85.864%	0.5710	120.0 min
Frozen frontier, online full fit	16	75.533%	0.9967	76.418%	1.0063	43.4 min
Single-layer pre-classifier integrator, adaptive LoRAs (full fit)	3	79.239%	0.8907	79.895%	1.0568	116.0 min

Maximum accuracy and its NLL are reported separately to expose calibration tradeoffs. Every cell also includes all 367 current-task identities; maximum historical H=11,827 is exactly the 12,194-image full fit.

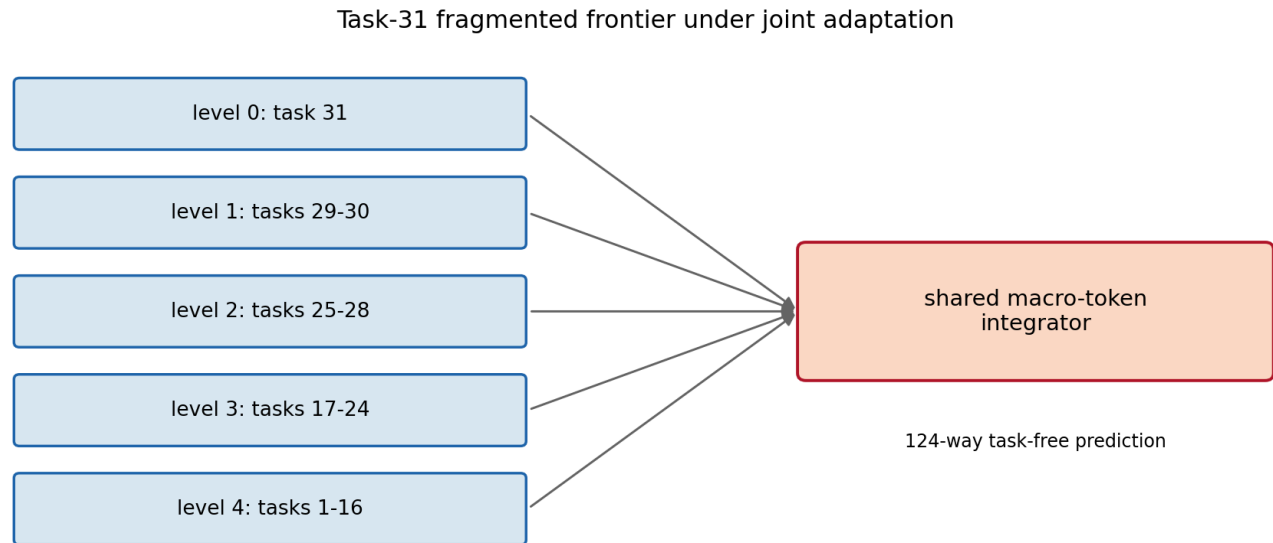
Joint-IID capacity trajectories

Epoch	Rank-80 accuracy	Rank-80 NLL	Rank-224 accuracy	Rank-224 NLL
1	75.992%	0.9977	79.272%	0.8326
2	78.485%	0.8350	81.109%	0.7311
3	80.026%	0.7954	81.699%	0.7199
4	80.354%	0.8340	82.060%	0.7727
5	80.125%	0.8339	81.109%	0.7772

Epoch five is the predeclared comparison endpoint. Minimum-NLL epochs are retained only as convergence diagnostics.

Architecture and experimental boundary

The task-31 frontier has five nodes at levels 0-4, covering task intervals 31, 29-30, 25-28, 17-24, and 1-16. Each node supplies its own final 197×768 LoRA-adapted token sequence and immutable local affine scores to the macro transformer. The new condition instead concatenates only the five 768-value pre-classifier outputs and maps the resulting 3,840 values directly to 200 logits. Neither head receives task IDs or labels.



Each arrow carries 197 adapted 768-value tokens plus local affine scores.

This diagram shows the macro condition: the base ViT and node classifiers are frozen, while five rank-16 LoRAs and the shared 12.06M-parameter macro head can move. The single-layer condition replaces that head with one affine map.

Every cell includes all 367 current-task images. $H=1,024, 2,048, 4,096, 8,192,$ and $11,827$ are nested prefixes of one deterministic uniform draw from the historical tasks without replacement. Maximum H therefore gives exactly all 12,194 fit identities. Every cell starts independently from identical sealed node tensors and macro initialization. The validation partition has 3,049 clean identities and never contributes gradients. No test image is opened.

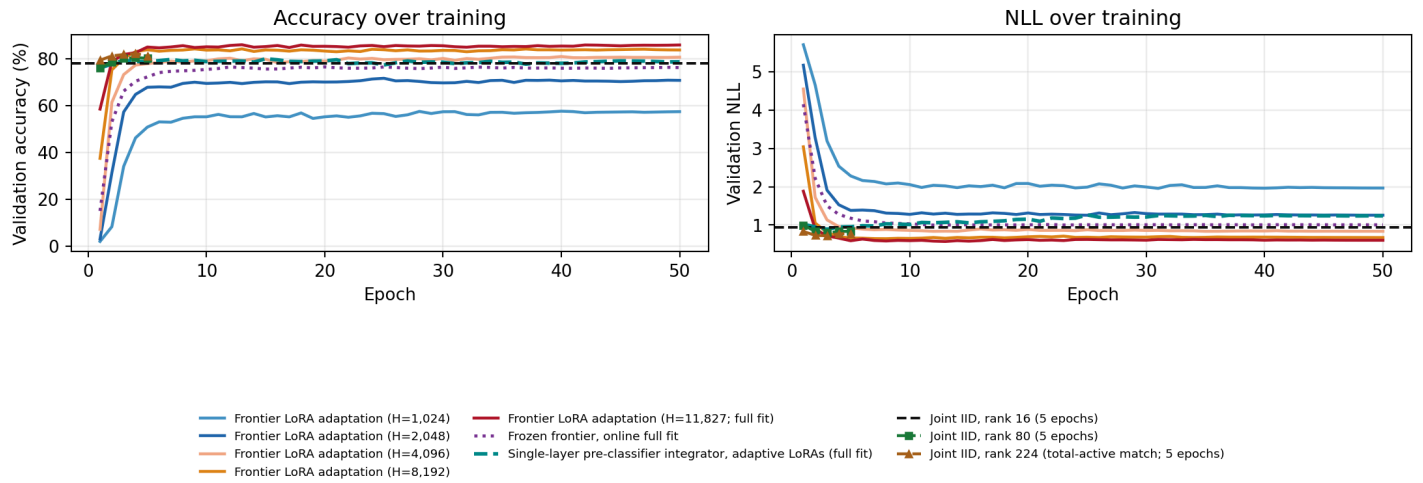
Compute boundary. Both adaptive frontier heads evaluate the same five node-specific ViTs for every image. The linear condition then runs one 768,200-parameter affine map; the macro condition runs its 12,055,496-parameter transformer. Joint IID evaluates one ViT with one shared LoRA and classifier. Their split and full-fit data exposure match, but deployment compute remains different.

Why the frozen online control exists

The previous frozen macro was trained from cached center-crop tokens. This run loads images online with deterministic random training augmentation. The purple full-fit control follows that new path while freezing the LoRAs, so its difference from the cached gray reference measures the pipeline/augmentation change rather than representation adaptation.

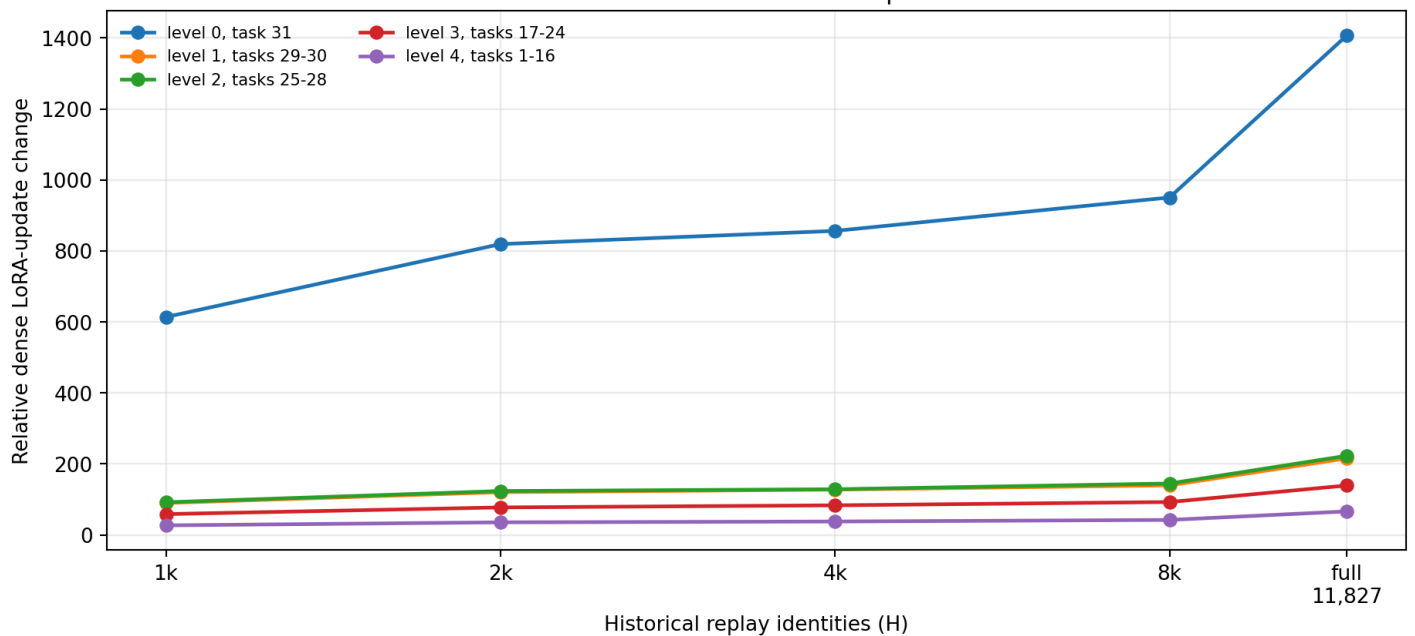
Optimization behavior

Frontier learning curves and joint-IID five-epoch controls



Every macro-frontier and single-layer epoch and all five epochs for both joint-IID capacity controls are retained in hash-chained histories. The horizontal black line is the rank-16 epoch-five endpoint, not a stopping gate.

How far each sealed frontier adapter moved



Macro-condition H sweep: scale-aware Frobenius movement of each dense LoRA update, measured from its sealed source node at the selected minimum-NLL checkpoint.

Interpretation

Allowing the frontier LoRAs to move changes the result by 10.331 percentage points and 0.4257 NLL relative to the online frozen-LoRA control at their minimum-NLL checkpoints. $H=4,096$ is the smallest tested historical population to cross both rank-16 joint-IID values. At exactly 5 full-fit passes (60,970 image presentations), adaptive full history is 84.880% / 0.5897, frozen full history is 72.155% / 1.1788, and rank-16 joint IID is 77.862% / 0.9425. This isolates feature adaptation from training exposure, but not the frontier's extra model and deployment compute.

What the single-layer condition establishes

The single-layer pre-classifier integrator reaches **79.239% accuracy / 0.8907 NLL** at its minimum-NLL checkpoint, epoch 3. Relative to the macro-token full-history checkpoint selected by the same rule, this changes accuracy by -6.625 points and NLL by +0.3197. At epoch five it reaches 78.649% / 0.9560, changing accuracy by -6.232 points and NLL by +0.3664 at identical exposure.

The head is one affine map from five concatenated 768-value pre-classifier vectors to 200 logits. It starts as the exact union of the frozen local classifiers and can then learn cross-node weights. Its 768,200 parameters are 93.6% fewer than the macro head's 12,055,496. Both conditions start from the same five source LoRAs and use the same data, augmentation, optimizer schedule, and checkpoint rule.

What the capacity controls establish

The rank-80 joint-IID control reaches **80.125% accuracy / 0.8339 NLL** at the fixed epoch-five endpoint: +2.263 accuracy points and -0.1085 NLL versus rank 16. The frontier remains 4.756 points higher and 0.2443 NLL lower.

Rank 80 and the frontier each expose 6,635,520 LoRA parameters and 60,970 image presentations. The frontier additionally trains a 12,055,496-parameter macro head, starts from five pretrained adapters, and executes five ViT paths; rank 80 starts one zero-effect adapter and uses one path. Its diagnostic minimum NLL is 0.7954 with 80.026% accuracy at epoch 3.

The rank-224 control reaches **81.109% accuracy / 0.7772 NLL** at epoch five. Its 18,674,812 trainable parameters are 16,204 (0.087%) fewer than the adaptive frontier's 18,691,016, which is the closest possible integer-rank match. Relative to rank 80, it changes accuracy by +0.984 percentage points and NLL by -0.0568. At matched exposure, the adaptive frontier remains 3.772 accuracy points higher, while the adaptive frontier is 0.1875 NLL lower. Its minimum NLL is 0.7199 at epoch 3; accuracy peaks at 82.060% at epoch 4. Validation NLL then rises by 0.0573 through epoch five while the training objective keeps improving, a pattern consistent with late overfitting or miscalibration under the inherited rank-16 schedule.

This removes active parameter count as a large numerical mismatch, but it does not match factorization, initialization, training history, or inference compute. Rank 224 is one fresh adapter in one ViT path; the frontier starts from five pretrained adapters and combines five ViT paths with a macro transformer.

The head schedule is the previous minimum-NLL winner: effective batch 64, peak AdamW LR $3e-5$, 50 epochs, 5% warmup, and cosine decay. The LoRA peak LR $5e-4$ is imported from the same-split joint recipe but run here in AdamW under the shared schedule. It has not been tuned for this coupled model.

Limits. This is a one-seed validation screen. The H cells change both unique data and total optimizer updates. Maximum accuracy is exploratory because all 50 validation checkpoints are visible. A promising condition needs replication and a focused LoRA/head learning-rate audit before any locked-test use.

Integrity and reuse

The training seals record zero fit-validation overlap, zero test overlap, zero test evaluations, and unchanged source hierarchy files. Fresh-process replay authenticated all six macro-frontier cells, the single-layer condition, and the rank-80 and rank-224 controls without taking an optimizer step. Large checkpoints and model weights remain local; compact histories, tables, plots, and protocol records are the report surface.